DSM Partitioning

A Pipeline for Reordering Dependency Structure Matrices

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Outline

Motivation

The challenge:

- Modern software & engineering projects involve **hundreds to thousands of interdependent tasks** across diverse teams
- Traditional tools (Gantt, CPM) handle sequential & parallel flows well...
- ...but **fail for interdependent tasks** where outputs feed back as inputs
- These feedback dependencies are common in complex product development

Goal

Find a **task ordering** that minimises conflicts in information flow to enable faster, more predictable project execution.

Key observation

When task j supplies input to task i but is scheduled *after* it, assumptions must be made. If wrong, task i must be **re-executed**.

Design Structure Matrix (DSM)

Definition: A binary matrix $A \in \{0,1\}^{n \times n}$ where

$$a_{ij} = 1 \iff \text{task } i \text{ depends on task } j$$

- Rows and columns labeled by the same task list
- Diagonal entries carry no dependency information.
- Equivalent to directed graph $G(A) = (V, E)$

Three dependency types:

- **Feed-Forward** ($j < i$): below diagonal — easy
- **Feed-Back** ($j > i$): above diagonal — problematic
- **Coupled**: mutual dependency — hardest

The Partitioning Problem

Objective: Find $P \in \{0,1\}^{n \times n}$ such that $A' = P^\top AP$ minimises:

Objective 1 — Feed-Back Marks

$$|\text{FBM}| = \left|\{a'_{ij} \neq 0 \mid j > i\}\right|$$

Minimise the **count** of above-diagonal nonzeros.

Objective 2 — Feedback Distance

$$\text{TFBD} = \sum_{a'_{ij} \in \text{FBM}} (j - i)$$

Minimise the **distance** of feed-back marks from the diagonal.

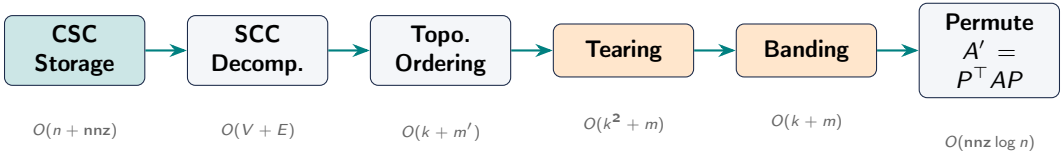
Best case

$G(A)$ has no directed cycles $\Rightarrow$ fully lower triangular,
$|\text{FBM}| = 0$, $\text{TFBD} = 0$

General case

Achieve **Block Lower Triangular Form (BLTF)**: group mutually dependent tasks into diagonal blocks and push remaining feed-back marks close to the diagonal.

Our 6-Stage Pipeline



Data Structure: CSC

- Column pointer array $\texttt{col\_ptr}[0 \ldots n]$
- Row index array $\texttt{row\_ind}[0 \ldots \text{nnz} - 1]$
- Storage: $\Theta(n + \text{nnz})$
- No value array needed (binary DSM)

Why CSC?

Column j gives the neighbours of vertex j in $O(\deg(j))$ — ideal for DFS traversal.

SCC Decomposition

Both algorithms implemented as **iterative DFS** over CSC, using an explicit stack (no recursion).

Tarjan's Algorithm

- **Single DFS pass**
- Each vertex v maintains $low(v)$: smallest reachable pre-order time
- $low(v) = v.pre$ at finish $\Rightarrow$ v is SCC root; pop stack
- $O(V + E)$ time, $O(V)$ space

Kosaraju-Sharir Algorithm

- **Two DFS passes**
- Pass 1: DFS on G, record finish order
- Pass 2: DFS on $rev(G)$ in reverse finish order; each DFS tree = one SCC
- $O(V + E)$ time, $O(V + E)$ space

After SCC decomposition, the **condensation DAG** is topologically sorted with **Kahn's algorithm** to determine the block ordering.

Tearing & Banding

Tearing (reduces FBM within SCCs)

- Reorder tasks within each SCC to reduce feed-back marks
- Uses **Algorithm GR**: greedily removes sinks, sources, then the vertex with the largest degree difference
- Backward edges in the resulting ordering are the "torn" feedback arcs

Banding (reduces TFBD within SCCs)

- Push remaining feed-back marks closer to the diagonal
- **Reverse Cuthill-McKee (RCM)**: concentrates nonzeros near diagonal
- **Adjacent-swap refinement**: accepts swaps that lower TFBD without increasing FBM
- Best of torn + RCM orderings is kept

Experimental Setup

Benchmark suite

- 10 sparse matrices from the project repository
- Dimensions: $n = 11$ to 27,770, nnz $= 4$ to 352,807

Hardware & methodology

- Apple M2 / 8 GB RAM (L1d 64 KB, L1i 128 KB, L2 4 MB), compiled with -O3, Apple LLVM 17
- Each algorithm: 2 warm-up + 5 timed runs
- Metrics: FBM count, TFBD, and mean runtime (μs)

Stage-wise Quality Improvement

Average reductions across 10 matrices:

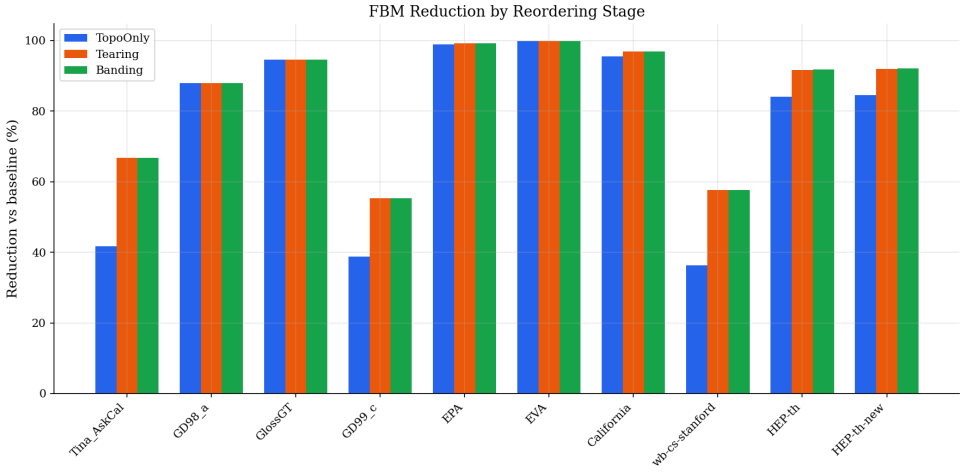
Stage	FBM ↓	TFBD ↓
Baseline	—	—
TopoOnly	76.18%	95.21%
+ Tearing	84.12%	93.95%
+ Banding	**84.15%**	**94.54%**

- Topo ordering provides the **largest single gain**
- Tearing drives further FBM reduction within SCCs
- Banding polishes TFBD without hurting FBM

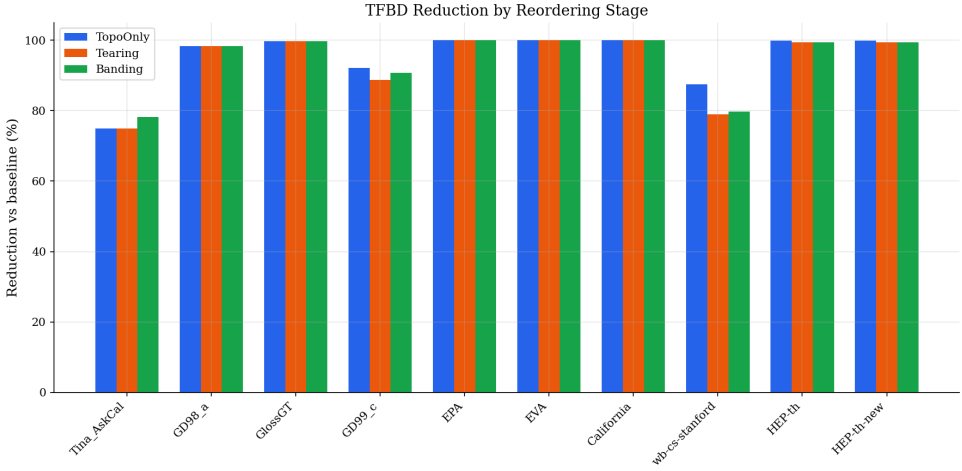
Selected matrix results

Matrix	$FBM_0 \rightarrow FBM_f$	$TFBD_0 \rightarrow TFBD_f$
Tina_AskCal	$12 \rightarrow 4$	7
California	$6063 \rightarrow 192$	669
wb-cs-stanford	$20131 \rightarrow 8537$	1.78M
HEP-th-new	$104435 \rightarrow 8281$	10.1M

FBM Reduction



TFBD Reduction



Runtime Comparison: Tarjan vs. Kosaraju-Sharir

Key findings

- Both algorithms produce **identical SCC decompositions**
- Tarjan: mean $\approx 560\,\mu$s
- Kosaraju-Sharir: mean $\approx 992\,\mu$s
- Tarjan is $\approx 1.77\times$ faster on average
- Scaling compatible with expected $O(V + E)$ near-linear behaviour

Why is Tarjan faster?

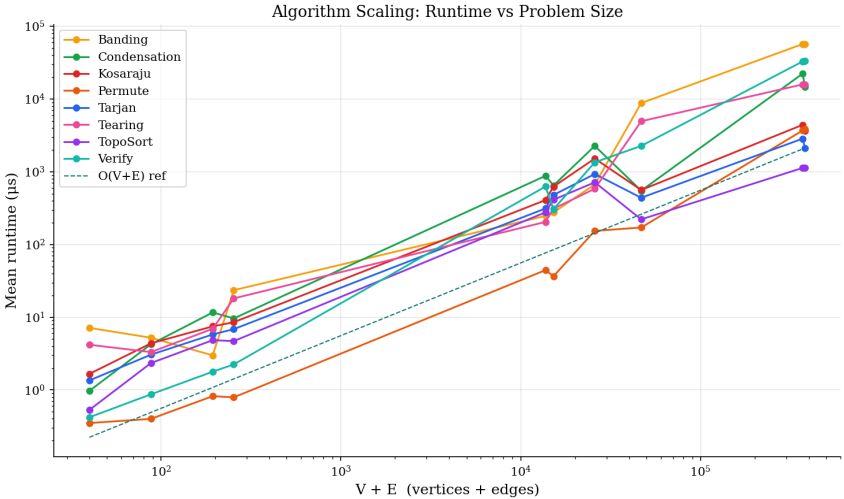
Single DFS pass vs. two passes + transpose graph construction in Kosaraju-Sharir.

Pipeline stage runtimes

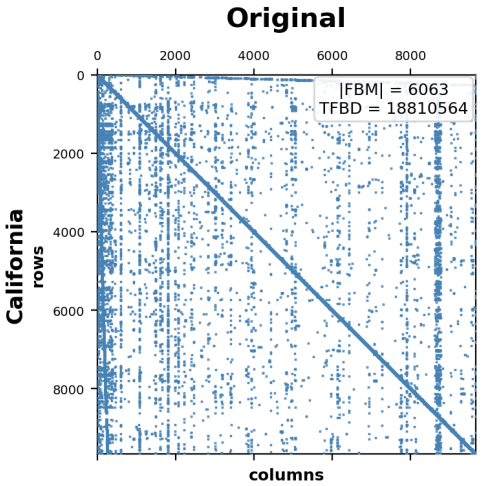
Stage	Relative cost
Tarjan SCC	low
Kosaraju SCC	$1.77\times$ Tarjan
Condensation	low
Topo sort	low
Tearing	moderate
Banding	highest
Permutation	low

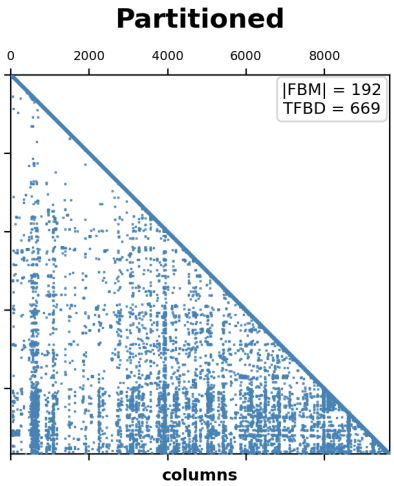
Banding is the heaviest stage but delivers the final quality gains.

Runtime Comparison: Pipeline

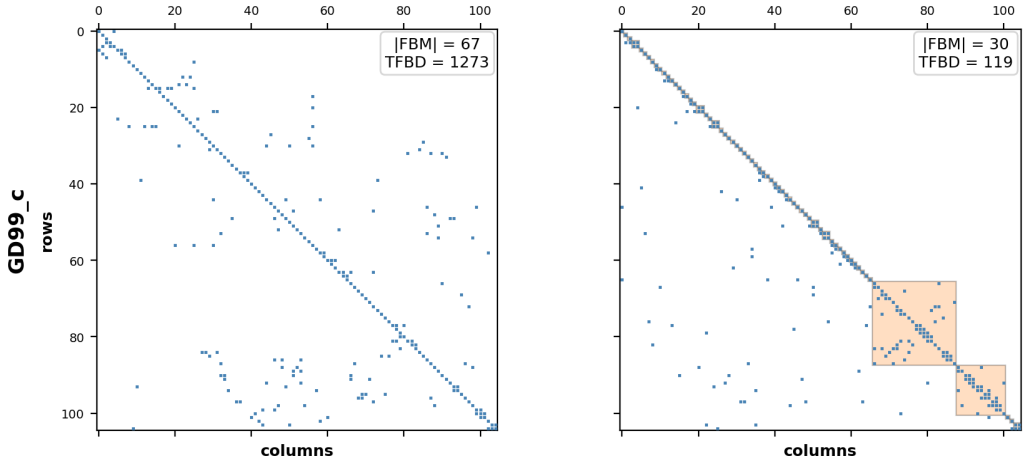


Qualitative Results: California

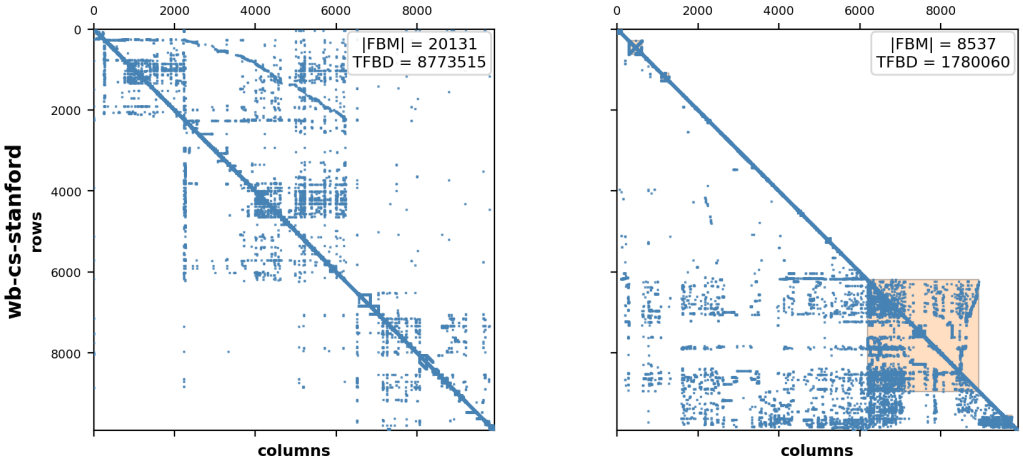




Qualitative Results: GD99_c



Qualitative Results: wb-cs-stanford



Conclusions & Future Work

What we achieved

- Implemented full 6-stage DSM reordering pipeline in C++ with CSC
- Both Tarjan and Kosaraju-Sharir SCC algorithms — identical decompositions
- **76.50% average FBM reduction**, **85.95% average TFBD reduction** across 10 benchmark matrices
- Tarjan 1.77× faster than Kosaraju-Sharir
- Runtime scales near-linearly with $V + E$

Future directions

- **Better tearing**: replace greedy Algorithm GR with an exact or improved heuristic for minimum feedback arc set
- **Parallelism**: per-SCC tearing & banding are independent — parallelise for large matrices
- **Weighted DSMs**: extend to numerical dependency strengths; optimise weighted feedback cost

Acknowledgements & Team Contributions

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Team Member Contributions

Member	Contribution
Simon Kraft	[placeholder]
Joshua Payne	[placeholder]
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Thank you — questions welcome!